

Peng Zhang

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EDUCATION

University of Wisconsin-Madison (UW-Madison), Madison, US Sep 2025 - Present
PhD in Electrical and Computer Engineering

- **GPA:** 4.0/4.0, **Advisors:** Prof. Manish Singh and Prof. Line Roald
- **Research:** Measurement Site Selection for Distribution Systems Monitoring and Data Centers Integration

University of Washington (UW), Seattle, US Sep 2023 - Jun 2025
MS in Electrical Engineering

- **GPA:** 3.95/4.0, **Advisor:** Prof. Baosen Zhang
- **Research:** Online Optimal Voltage Control in Power Distribution Networks

North China Electric Power University (NCEPU), Beijing, China Sep 2019 - Jun 2023
BEng in Electrical Engineering and Its Automation

- **GPA:** 93.08/100 (3.95/4.0), **Ranking in 3 years:** 1/55, 1/399, 7/396
- **Thesis:** High-Resolution Spatiotemporal Assessment of Onshore and Offshore Wind Energy and Long-Term System Planning, advised by Prof. Nian Liu

Yale University, Online Jul 2022 - Aug 2022
Summer session in Green Energy Systems

- **Final report:** Life Cycle Impact Assessment (LCIA) of Hydrogen Production Pathways, advised by Prof. Yehia Khalil

PUBLICATIONS

- [1] **P. Zhang**, L. Roald and M. Singh. "Optimal Measurement Selection for Certifiable Voltage Monitoring in Power Distribution Systems," accepted to HICSS-60. [[Poster](#)]
- [2] **P. Zhang**, L. Roald and M. Singh. "A Characterization of Measurement Informativity for Voltage Monitoring in Distribution Networks," (In progress)
- [3] **P. Zhang** and B. Zhang, "Optimal voltage control using online exponential barrier method," in *Proc. 57th North American Power Symposium (NAPS)*, 2025. [[Slides](#)]
- [4] **P. Zhang** and B. Zhang, "Online voltage regulation of distribution systems with disturbance-action controllers," *arXiv preprint arXiv:2412.00629*, 2024. Appeared as poster at PESGM 2025. [[Poster](#)]
- [5] Y. Zhang, L. Wang, **P. Zhang**, H. Luo, W. Shi, and X. Wang, "The nonlinear wave solutions and parameters discovery of the Lakshmanan-Porsezian-Daniel based on deep learning," *Chaos, Solitons & Fractals*, vol. 159, Art. no. 112155, 2022.

RESEARCH EXPERIENCE

Online Voltage Control for Distribution Systems under Model Inaccuracy Jan 2024 - Aug 2025
Research Assistant; Supervisor: Prof. Baosen Zhang, UW

- **Real-time OPF:** Designed iterative algorithms for both active and reactive power injections of the inverters, in contrast to the offline OPF problem resolved in one shot; utilized the real-time feedback from distribution networks for updating control inputs to handle model inaccuracies
- **Disturbance-action controller (DAC):** Explicitly leveraged the correlation of uncontrollable loads and regarded them as system disturbances; formulated the feedback controller - DAC, to handle communication delay and time-varying load and solar information; derived the closed-form expression for the closed-loop system dynamics and provided theoretical results on stability and performance degradation under inaccurate models
- **Online exponential barrier:** Proposed a barrier-based Lagrangian method for a better trade-off between safety and optimality of voltage control under model inaccuracies, compared to the classic barrier method (double-loop) and Lagrangian (linear addition); improved intermediate voltage constraint satisfaction; provided the analytical selection rule for barrier weights and proved the safety guarantee at convergence

Spatiotemporal Modeling and Planning of Large-scale Power Systems

Sep 2022 - Jun 2023

Research Assistant, Supervisor: Prof. Nian Liu, NCEPU

- **Resources assessment:** Evaluated physical potential of PV and wind energy by processing hyperscale and high-resolution numerical weather and geospatial data from Seniverse, NREL, NASA, etc., based on PVLIB and a modified wind output model; assessed economic potential by calculating the Levelized Cost of Electricity (LCOE)
- **System planning:** Formulated the large-scale linear/quadratic programming for the optimal system expansion and operation scheme within provincial grids with integration of utility-scale PV, wind, energy storage, etc.
- **Adequacy under extreme weather:** Characterized the nonlinear response of loads to the outdoor temperature; served as the team member to comprehensively evaluate the nationwide adequacy and flexibility of renewable energy during diverse extreme weather events, e.g., hot and cold waves. [[Project Paper](#)]

Interdisciplinary Research on Machine Learning and Nonlinear Dynamics

May 2020 - Jun 2023

Research Assistant; Supervisor: Prof. Lei Wang, NCEPU

- **Analytical methods:** Applied Linear Stability Analysis (LSA), Darboux Transformation (DT) to explore the evolution of solution dynamics for governing high-order and high-dimensional nonlinear PDEs
- **Data-driven methods:** Leveraged Physics-Informed Neural Network (PINN) to both discover the mechanism of control equations (backwards) and infer the solutions (forwards) under corresponding conditions; extended PINN to (2+1)-d in the context of DS I equation that describes the wave dynamics in shallow water; predicted the evolution of rogue waves and explored the transition conditions among three families of lump waves

INTERNSHIP**GE Vernova**, Bothell, WA

Jun 2024 - Aug 2024

Software Engineer Intern – Power Systems

- Gained hands-on experience with GE Vernova's software *Platform and Habitat* delivered to U.S. utilities for analysis and operation of the transmission system
- Developed an optimal scheduling method for controllable reactive power resources, including automatic voltage regulators (AVRs) and shunt capacitor banks, to minimize transmission system losses; derived the underlying analytical relationships and implemented the framework as a Python module for integration into existing Energy Management System (EMS) software
- Solved for cost-effective actions to satisfy utility-specific voltage requirements and applied chance-constrained optimization to improve system robustness against large uncertainties from renewable energy resources

AWARDS & HONORS

Novotny Welcome Awards, UW-Madison

Oct 2025

NAPS Student Travel Award, Hartford CT

Oct 2025

IEEE PES GM Student program, Austin TX

Jul 2025

Outstanding Graduated Undergrad in Beijing (4%), NCEPU

Jun 2023

Extra-high Voltage (EHV) Scholarship (0.4%), NCEPU

Sep 2022

First-class Schneider Electric Scholarship (1%), NCEPU

Sep 2021

First-class Academic Scholarship (5%), NCEPU

Sep in 2020, 2021, 2022

COMPETITION**Mathematical Contest in Modeling (MCM)**

Feb 2021

Team Leader; Finalist Winners, Top 2% Worldwide

- Formulated a framework of wildfire risk assessment and resource allocation using NASA climate data.
- Adapted the ID3 decision tree to classify wildfire risk across discretized geographic regions, applied time-series analysis (TSA) to forecast fire occurrence, and derived a data-driven drone procurement strategy for rapid wildfire response in Australia.

EXTRACURRICULAR ACTIVITIES

Volunteer for Discovery Day 2025, UW

May 2024

- Guided 4th–8th grade students and their parents in hands-on science exhibition activities and promoted engagement within the interdisciplinary engineering community.

Captain of Soccer Team, Department of Mathematics and Physics at NCEPU

Sep 2020 - Jun 2021

- Took charge of the annual recruitment, weekly training, match deployment, etc.

SKILLS

Programming: Python, MATLAB & Mathematica (Proficient), C, Julia (Moderate)

Tools: (*Optimization*) Gurobi/Mosek/CPLEX, CVX; (*Power Systems*) MATPOWER/Pandapower/PyPSA, Multisim, PVLIB, PowerWorld, COMSOL; (*Machine Learning*) PyTorch and TensorFlow